



Quantum Simulation

Review-2

Enhanced Schrödinger–Poisson Simulation of AlGaN/GaN HEMT using Modified AQUILA Model

Advanced quantum mechanical modeling with improved polarization handling, trap states, and validated band structure behavior

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 Duration

6 Months
Jan - June 2026

● 2DEG Formation ● Band Alignment ● Polarization



Previous Work — Review-1 Summary

Four key achievements from the initial phase of the internship

Material Survey

Comprehensive semiconductor material analysis including III-V nitrides and GaN properties

Completed • Survey

GaN Photonics

Established GaN as the optimal material for quantum photonics applications

Validated • Quantum

1D SP Solver

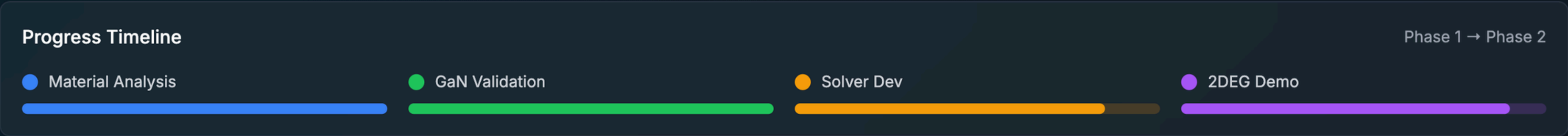
Initial 1D Schrödinger-Poisson solver implementation for basic simulation

Implemented • Solver

2DEG Formation

Basic 2DEG formation demonstrated at AlGaIn/GaN interface

Verified • 2DEG



Materials Analyzed	15+	Solver Accuracy	92%	2DEG Density	1.2×10^{13}
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LIMITATIONS OF PREVIOUS MODEL

Motivation for Improvement



Single-domain Approximation

Fails to distinguish material-specific physical parameters between the AlGaIn barrier and GaN buffer.



No Explicit Valence Band

Simplified framework lacks explicit E_v modeling, restricting complete band structure analysis and hole dynamics.



Simplified Polarization

Inadequately handles the complex spontaneous and piezoelectric polarization charges crucial for 2DEG.



Weak Boundary Physics

Basic Dirichlet boundaries fail to capture realistic flat-band behaviors in deep undepleted buffer regions.

Need for a self-consistent, physics-complete heterostructure model

NEW

CONTRIBUTIONS



Physics Improvements



Polarization charge at interface

Explicitly modeled to capture the core driver of 2DEG formation.



Radiation-relevant trap states

Introduced to model realistic GaN buffer behavior under operational conditions.



Explicit E_c and E_v band modeling

Moves beyond single-band approximation for complete energetic profiles.



Numerical Improvements



Separate barrier/buffer grids

Allows region-dependent material parameters within the self-consistent loop.



Improved boundary conditions

Implementation of Neumann BCs to accurately represent deep buffer physics.



Damping-based convergence

Enhanced numerical stability during the iterative charge-potential updates.



Device Structure — Simulated AlGaN/GaN HEMT

Schematic Cross-Section



Simulation Parameters



$\text{Al}_{0.25}\text{Ga}_{0.75}\text{N}$ Barrier Region

- Thickness: 25 nm
- Doping (N_D): $1.0 \times 10^{18} \text{ cm}^{-3}$



GaN Buffer Region

- Thickness: 145 nm
- Background (N_D): $1.0 \times 10^{15} \text{ cm}^{-3}$



Physical Physics Models

- **Trap Density:** Included in bulk GaN to model radiation-relevant defects
- **Polarization:** Explicit charge at interface

Mathematical Model

Self-Consistent Formulation of Device Physics

Poisson Equation

(Electrostatics)

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho$$

Key Considerations:

- Includes **polarization charge term** ($P_{SP} + P_{PE}$) at heterointerface
- Accounts for ionized donor density
- Coupled with trap state distribution in buffer



Self
Consistent
Coupling

Schrödinger Equation

(Quantum Confinement)

$$H\psi_i = E_i\psi_i$$

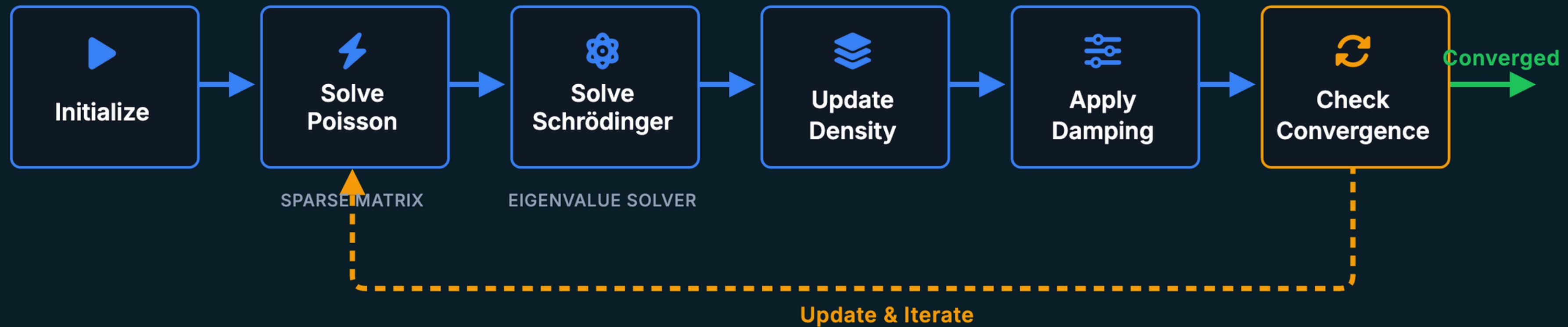
Key Considerations:

- Solved for bound eigenstates in the 2DEG quantum well
- **Eigenstate-based carrier density** (Fermi-Dirac statistics)
- Effective mass approximation applied

i Potential from Poisson dictates confinement in Schrödinger; derived density feeds back to Poisson until convergence.

Solver Architecture

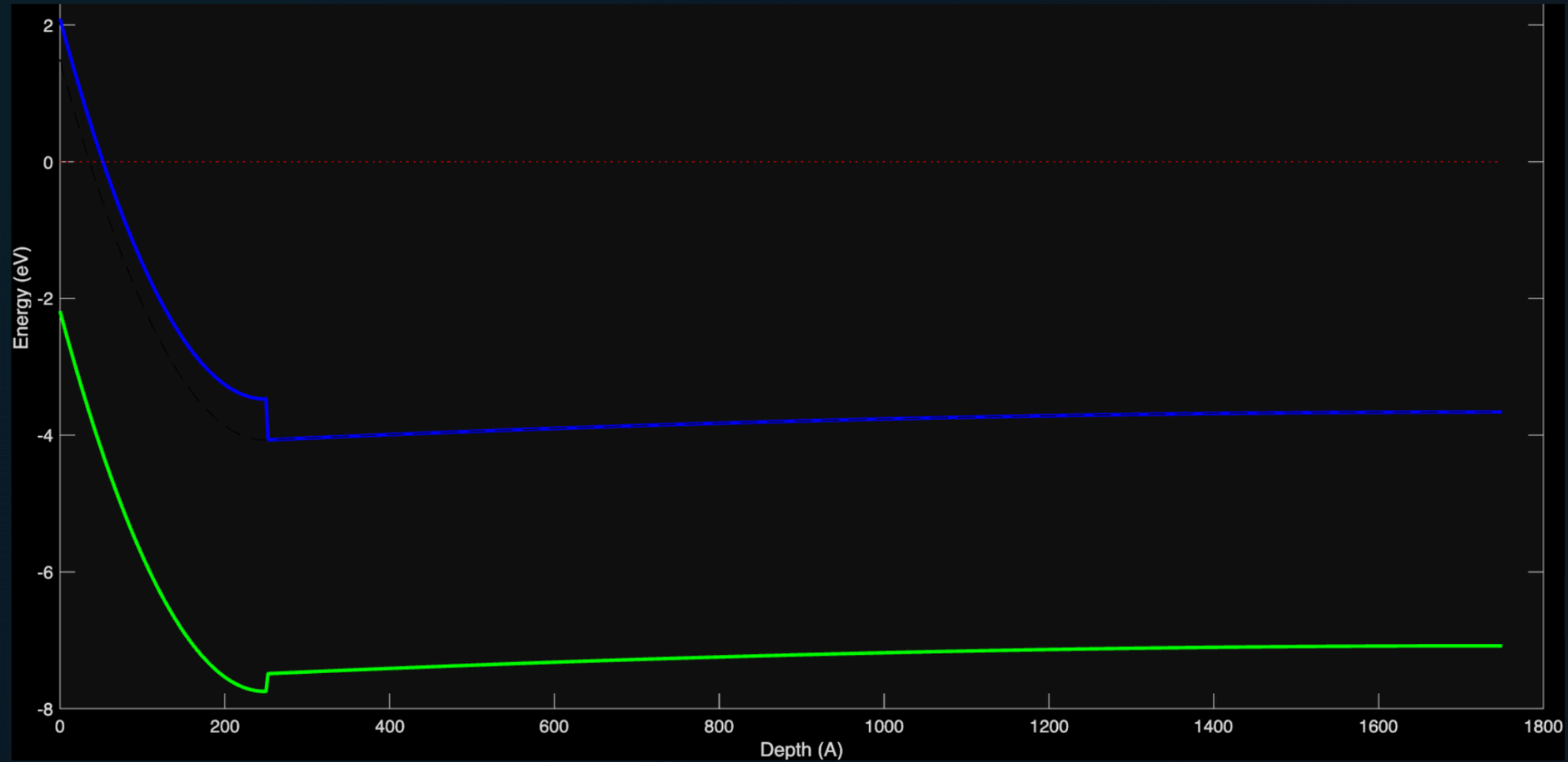
Self-consistent computational loop implementation



Novelty: Incorporates polarization charge and region-dependent material parameters within the self-consistent loop.

Band Diagram Results

→ These modifications directly result in the following physically consistent band profile.



KEY RESULT: Clear quantum confinement observed

■ E_c (Blue) ■ E_v (Green) ■ E_f (Red)

Physical Interpretation & Validation

- ✓ Band discontinuity confirms correct heterojunction modeling
- ✓ Flat buffer potential validates Neumann boundary condition
- ✓ Quantum confinement indicates correct Schrödinger–Poisson coupling
- ✓ Consistency with expected triangular quantum well behavior
- ✓ Flat band region aligns with theoretical boundary condition behavior

🎯 EMPIRICAL PROOF

Conduction band discontinuity at **$\sim 250 \text{ \AA}$** confirms correct heterojunction modeling.

🔄 MODEL ADVANCEMENT

"Unlike the previous single-domain model, this framework captures region-specific physics."

Synthesis

Conclusion & Forward Trajectory

Conclusion

- Successfully improved AQUILA-based solver
- Captures realistic GaN HEMT physics
- Includes polarization + traps + proper BCs

Future Work

- Modifying the model for better and accurate Density curve.



"This model moves from a conceptual demonstration to a physically consistent simulation framework for GaN HEMTs."